

CIRCULAR LINKED LIST

PROGRAM CODING:

```
#include<stdio.h>

#include<malloc.h>

struct node

{

    int info;

    struct node *next;

}*h;

void create(struct node *);

void insertfront(struct node *,int);

void insertmiddle(struct node *,int);

void insertend(struct node *,int);

void deletefront(struct node *);

void deletemiddle(struct node *);

void deleteend(struct node *);

void search(struct node *);

void display(struct node *);

void main()

{

    struct node *h;

    int n,s,op;

    clrscr();

    h=(struct node *)malloc(sizeof(struct node *));
```

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printf("\n\n\n\t\t\t Linked list operation");

printf("\n\n\t1.Create\n\t2.Insert front\n\t3.Insert middle\n\t4.Insert end\n");

printf("\n\t5.Delete\t\t\t\t\t front\n\t6.Delete\t\t\t\t\t middle\n\t7.Delete
end\n\t8.search\n\t9.Display\n\t10.Exit\n");

while(1)
{
    printf("\n Enter your option:\t");

    scanf("%d",&op);

    switch(op)
    {
        case 1:create(h);break;

        case 2:

            printf("\n Enter the number to insert:\t");

            scanf("%d",&n);

            insertfront(h,n);

            break;

        case 3:

            printf("\n Enter the number to insert:\t");

            scanf("%d",&n);

            insertmiddle(h,n);

            break;

        case 4:

            printf("\n Enter the number to insert:\t");

            scanf("%d",&n);

            insertend(h,n);

            break;

        case 5:
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        deletefront(h);

        break;

    case 6:

        deletemiddle(h);

        break;

    case 7:

        deleteend(h);

        break;

    case 8:

        search(h);

        break;

    case 9:

        display(h);

        break;

    case 10:

        exit(0);

        break;

    default:printf("\n.....Invalid option\n");

}

}

}

void create(struct node *h)

{

    int i,n,x;

    struct node *s1,*s2;

    s1=(struct node *)malloc(sizeof(struct node *));

    printf("\n Enter the number of elements:\t");

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scanf("%d",&x);

printf("\n Enter the element:\t");

scanf("%d",&n);

s1->info=n;

s1->next=h;

h->next=s1;

for(i=0;i<x-1;i++)

{

    s2=(struct node *)malloc(sizeof(struct node *));

    s1->next=s2;

    printf("\n Enter the element:\t");

    scanf("%d",&n);

    s2->info=n;

    s2->next=h;

    s1=s1->next;

}

display(h);

printf("\n");

}

void insertfront(struct node *h,int n)

{

    struct node *x;

    x=(struct node *)malloc(sizeof(struct node *));

    x->info=n;

    x->next=h->next;

    h->next=x;

    display(h);

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    printf("\n");
}

void insertmiddle(struct node *h,int n)
{
    struct node *x,*y;
    int e;
    x=(struct node *)malloc(sizeof(struct node *));
    x->info=n;
    x->next=h;
    printf("\n Enter the element after which the given element to be inserted:\t");
    scanf("%d",&e);
    y=h;
    while(y->info!=e)
    y=y->next;
    x->next=y->next;
    y->next=x;
    display(h);
}

void insertend(struct node *h,int n)
{
    struct node *x,*y;
    x=(struct node *)malloc(sizeof(struct node *));
    x->info=n;
    x->next=h;
    y=h;
    while(y->next!=h)
    y=y->next;

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y->next=x;
display(h);
}
void deletefront(struct node *h)
{
    if(h->next!=h)
        h->next=h->next->next;
    else
        h=h;
    display(h);
}
void deletemiddle(struct node *h)
{
    int e;
    struct node *x,*y;
    y=h;
    x=y->next;
    printf("\n Enter the element \t");
    scanf("%d",&e);
    while(x->info!=e)
    {
        x=x->next;
        y=y->next;
    }
    y->next=x->next;
    display(h);
}

```

```

void deleteend(struct node *h)
{
    struct node *x,*y;
    y=h;
    x=y->next;
    while(x->next!=h)
    {
        x=x->next;
        y=y->next;
    }
    y->next=h;
    display(h);
}

void search(struct node *h)
{
    struct node *x,*y;
    int s;
    printf("\n Enter the number:\t");
    scanf("%d",&s);
    y=h;
    while(y->info!=s&& y->next!=h)
    y=y->next;
    if(y->info==s)
    printf("Element found\n");
    else
    printf("Element not found \n");
}

```

```
void display(struct node *h)
{
    struct node *y;
    y=h;
    printf("h->");
    while(y->next!=h)
    {
        y=y->next;
        printf("%d->",y->info);
    }
    printf("h");
}
```